

Equality and Reduction

(Equality/Reduction Rule Definition)

Description: Defines a reduction rule for expressions. The left-hand side is a pattern, and the right-hand side is the result of the reduction.

Parameters:

- Pattern: The expression to be matched.
- Result: The expression to replace the matched pattern with.

Return: Not reduced itself unless custom equalities over equalities are added

Example:

```
(= (add 1 2) 3)
```

(Identity)

Description: Returns its argument unchanged.

Parameters:

- Input: Any atom.

Return: The input atom.

Example:

```
!(id 5) ; Returns 5
```

=alpha

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Description: Checks the alpha equivalency of the two given atoms.

Parameters:

- ExpressionAtom: First expression.
- ExpressionAtom: Second expression.

Return: True if the expressions are alpha-equivalence False otherwise wise.

Example:

```
!=(alpha (Father $Jhon) (Father $Bob)) ; Returns True because they differ only by variable.  
!=(alpha (Father $Jhon) (Son $Jhon)) ; Returns False because they differ structurally.
```

assertEqual

Description: Compares (sets of) results of evaluation of two expressions.

Parameters:

- ExpressionAtom: First expression.
- ExpressionAtom: Second expression.

Return: Unit atom if both expressions after evaluation are equal, error - otherwise.

Example:

```
!(assertEqual (+ 1 2) (- 6 3)) ; Returns ().  
!(assertEqual (+ 1 2) (+ 2 2))  
;(Error (assertEqual (+ 1 2) (+ 2 2))  
;Expected: [4]  
;Got: [3]  
;Missed results: 4  
;Excessive results: 3)
```

assertAlphaEqual

Description: Asserts the alpha-equivalency of the two given expressions.

Parameters:

- ExpressionAtom: First expression.
- ExpressionAtom: Second expression.

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Return: Unit atom if both expressions after evaluation are alpha equal, error - otherwise.

Example:

```
!(assertAlphaEqual (+ $x $y) (+ $a $b)) ; Returns ().
!(assertAlphaEqual (+ $x $y) (- $x $y))
; Returns:
;(Error (assertAlphaEqual (+ $x $y) (- $x $y))
;Expected: [(- $x $y)]
;Got: [(+ $x $y)]
;Missed results: (- $x $y)
;Excessive results: (+ $x $y))
```

assertEqualToResult

Description: Same as assertEquals but it doesn't reduce the second argument. Second argument is considered to be a set of values of the first argument's evaluation.

Parameters:

- ExpressionAtom: First expression.
- ExpressionAtom: Second expression.

Return: Unit atom if both expressions after evaluation of the first argument are equal, error - otherwise.

Example:

```
!(assertEqualToResult (+ 1 2) 3) ; Returns ().
!(assertEqualToResult (+ 1 2) (+ 1 2))
;Returns:
;(Error (assertEqualToResult (+ 1 2) (+ 1 2))
;Expected: [+, 1, 2]
;Got: [3]
;Missed results: +, 1, 2
;Excessive results: 3)
```

assertAlphaEqualToResult

Description: Same as assertEqualToResult but for assertAlphaEqual. Checks the alpha-equivalency without reducing the second parameter.

Parameters:

- ExpressionAtom: First expression.
- ExpressionAtom: Second expression.

Return: Unit atom if both expressions after evaluation of the first argument are equal, error - otherwise.

Example:

```
(= (add) $x)
!(assertAlphaEqualToResult (add) ($y)) ; Returns ().
!(assertAlphaEqualToResult (add) (add))
;Returns:
;(Error (assertAlphaEqualToResult (add) (add))
;Expected: [add]
;Got: [$X#66]
;Missed results: add
;Excessive results: $X#66)
```